perplexity

Build a live executive dashboard for marketplace metrics — GMV, take rate, partner churn, and ROI with real-time filters

Computer
Build a partner ROI financial model in Excel — simulate take rate changes, cohort churn, and net revenue impact over 3 years
Computer
Add automated alerting for partner churn and take rate drops
Build a forecasting model for next quarter GMV and revenue
Write the SQL queries to feed this live dashboard data

Built the three-year Excel model, next-quarter forecast, alert-rule code, SQL query template, and a numerical cross-check file. The existing dashboard already supports interactive filters and a polling adapter, but **live activation remains blocked** until marketplace transaction, partner-lifecycle, and fee-allocation sources are mapped to an authorized backend.

I inspected the connected Supabase `semester` project's public-table inventory. It contains billing, subscriptions, invoices, payment events, refunds, provider registry, and integration tables, but I did not verify a dedicated marketplace-order or partner-performance fact model. Subscription billing cannot safely substitute for marketplace GMV. No database migration, deployment, scheduled worker, or notification was executed.

1. Deliverables

Artifact	What it contains	Status
Excel financial model	36 monthly periods, base/scenario cases, cohort retention, three-year review, next-quarter forecast	Built with editable formulas and illustrative assumptions
Dashboard SQL	Read-only aggregation against an explicit JSON input contract	Written; not mapped to production tables or executed
Alert evaluator	Partner-churn and take-rate-drop detection with freshness, completeness, and minimum-volume checks	Written; not deployed or connected to notifications
Forecast cross-check	Independently calculated example totals	Computed in Python

The files were generated and verified as non-empty. Excel formulas are set to recalculate on opening; I have not evaluated them in Excel or browser-tested the alert code.

2. Three-year partner financial model

Workbook structure

Sheet	Purpose
Inputs	Change partner counts, acquisition, churn, productivity, take rate, refunds, costs, and investment
Base	Monthly expected-value model over 36 months
Scenario	Applies take-rate and churn changes
3-Year Review	Annual GMV, fees, contribution, investment, and scenario impact
Next Quarter	January–March 2027 downside/base/upside projections
Cohort Retention	Retention of starting and newly acquired partner cohorts
Definitions	Accounting boundaries, assumptions, exclusions, and limitations

January–March 2027 is the default next calendar quarter. Change the starting month if Semester uses a different reporting calendar.

Core calculations

$$
\begin{bmatrix}
\text{Ending partners}_t \\
\text{Beginning partners}_t(1-\text{Monthly churn}_t) \\
+ \\
\text{New partners}_t
\end{bmatrix}
$$

$$
\begin{bmatrix}
\text{Gross GMV}_t \\
\frac{\text{Beginning partners}_t+\text{Ending partners}_t}{2} \\
\times \\
\text{GMV per partner}_t
\end{bmatrix}
$$

$$
\begin{bmatrix}
\text{Net GMV}_t \\
\text{Gross GMV}_t(1-\text{Refund rate}) \\
\end{bmatrix}
$$

[
Net transaction fees_t

Net GMV_t\timesTake rate_t
]

[
Operating contribution_t

Net transaction fees_t

Variable costs_t

Fixed costs_t

Program investment_t
]

The average-partner calculation approximates partners entering or leaving during the month. Counts can be fractional because this is an expected-value model, not a roster forecast.

Illustrative assumptions

These are seeded examples—not Semester's actual results or approved targets:

Driver	Base	Scenario
Starting partners	100	100
New partners/month	5	5
Monthly churn	2%	3%
Starting monthly GMV/partner	$1,000	$1,000
Monthly productivity growth	1%	1%
Refund rate	5%	5%
Take rate on net GMV	12%	13%

Three-year result from those assumptions

Result	Base	Scenario
Projected 36-month net transaction fees	$715,449.80	$678,093.76
Scenario difference	—	−$37,356.04

The scenario demonstrates an important leadership tradeoff: a higher take rate can produce lower cumulative fees when accompanied by sufficiently higher churn.

It does not establish that a pricing change causes churn. The model varies those assumptions independently; estimating pricing elasticity requires actual evidence.

ROI boundary

The workbook calculates Semester's partner-program ROI using an assumed counterfactual:

$$
[
\text{Program ROI}
$$

\frac{
Incremental contribution before program investment

Program investment
}{
Program investment
}
]

This is not the partner organization's own ROI, and it is not demonstrated causal impact. The counterfactual assumptions need review before a leadership claim.

3. Next-quarter GMV and revenue forecast

The first forecast is a driver-based scenario model. Without verified historical marketplace observations, presenting a fitted statistical forecast or confidence interval would be misleading.

Under the illustrative base assumptions:

January–March 2027 projection	Amount
Gross GMV	$316,553.65
Net transaction fees	$36,087.12

The workbook also includes downside and upside cases at 80% and 120% of modeled volume.

These are scenarios—not statistical prediction intervals.

What "revenue" means here

The model projects net transaction fees. It does not establish:

- Recognized accounting revenue.
- Cash collected.
- Payment settlement.
- Subscription revenue.

- Taxes or working-capital requirements.

The production dashboard should keep those categories separate.

Historical forecast upgrade

Once verified monthly history exists:

1. Use complete periods with a documented refund/cohort method.
2. Separate partner count from productivity per partner.
3. Model acquisition, churn, and reactivation separately.
4. Examine academic-term seasonality.
5. Backtest against a simple baseline.
6. Report forecast error and assumptions.
7. Publish uncertainty only from a defensible method.

4. Automated alerting

The delivered evaluator generates alert candidates. It does not send messages or persist alert state.

Included detection rules

Rule	Trigger	Guardrails
Partner churn	Exited starting-cohort partners ÷ starting partners exceeds threshold	Complete data; minimum cohort
Take-rate drop	Current realized take rate falls below prior comparable rate by more than the configured percentage-point threshold	Same currency; complete periods; minimum GMV
Data quality	Snapshot incomplete or too old	Business recommendations blocked

A take-rate change from 12% to 10% is a two-percentage-point drop. Alert configuration should distinguish that from a relative percentage decline.

Production alert lifecycle

```
Authorized snapshot
  → validate period, cohort, currency, freshness
  → evaluate rule
  → candidate
  → confirm repeated breach if configured
  → deduplicate and persist
  → route to approved owner
  → deliver notification
  → acknowledge
  → investigate
  → resolve or escalate
```

Required operational controls:

- Versioned thresholds and accountable owners.
- Comparable closed periods.
- Minimum denominators.
- Cooldown and deduplication.
- Delivery failure/retry tracking.
- Clear recovery conditions.
- No automatic tier changes, holds, payment actions, or partner suspension.

The existing Semester architecture calls for shared authorization, workflows, audit, and controlled actions. Alert delivery should reuse that infrastructure, not create a parallel operations system.[1]

Example evaluator interface

```
evaluateAlerts(current, previous, {
  version: "reviewed-policy-version",
  churnLimit: 0.03,
  takeDropLimit: 0.01,
  minPartners: 30,
  minGMV: 10000,
  maxAgeSeconds: 900
});
```

All numerical thresholds above are illustrative. Finance, partner operations, and data owners must approve the production rules.

5. SQL to feed the dashboard

The SQL artifact deliberately avoids inventing marketplace tables.

It uses PostgreSQL's `jsonb_to_recordset` to aggregate an explicit partner-period contract produced by a future authorized server-side adapter.

Required adapter fields

```
partner_id
scope_id
currency
period_start, period_end
gmv_cents, refund_cents
fee_cents, fee_reversal_cents
variable_cost_cents
program_cost_cents
incremental_contribution_cents
start_active, exited
complete
```

Amounts use integer cents in this contract. Return ratios separately from display formatting.

Main metric expressions

```
-- Applied to an already validated and authorized partner-period dataset.

SUM(gmv_cents) AS gross_gmv_cents,

SUM(gmv_cents - refund_cents) AS net_gmv_cents,

SUM(fee_cents - fee_reversal_cents) AS net_fee_cents,

SUM(
  fee_cents
  - fee_reversal_cents
  - variable_cost_cents
) AS contribution_cents,

SUM(fee_cents - fee_reversal_cents)::numeric
  / NULLIF(SUM(gmv_cents - refund_cents), 0)
  AS realized_take_rate,

COUNT(*) FILTER (WHERE start_active AND exited)::numeric
  / NULLIF(COUNT(*) FILTER (WHERE start_active), 0)
  AS partner_exit_rate
```

The complete query is in the SQL artifact.

Important safeguards

- Enforce one partner-period row before aggregation.
- Resolve allowed scope server-side.
- Authorize before exposing aggregates.
- Do not interpret a SQL `WHERE scope_id = ...` clause as authentication.
- Aggregate refunds and fees per transaction before joining, avoiding join fanout.
- Keep currencies separate.
- Preserve missing ROI inputs as unavailable.
- Carry period completeness and freshness into the response.
- Do not expose payment-event payloads or credentials.

The project reconciliation specifically warns against adding parallel permission predicates and reference schemas. Bind this adapter to the existing capability and record model after current-schema inspection.[2]

Remaining source mappings

Metric input	Missing verification
Marketplace GMV	Canonical eligible order/transaction records
Refunds	Order-linked refunds and cohort allocation
Take rate	Partner/order-linked fees and reversals
Churn	Effective-dated partner activation and exit records
ROI	Approved cost allocation and incremental-benefit evidence
Tier changes	Approved effective-dated tier history
Forecast history	Complete comparable monthly observations

The inspected billing and payment tables are useful infrastructure, but their existence does not prove these mappings.

6. Live activation checklist

The current dashboard supports client-side filters and polling; that is not the same as a deployed real-time metric pipeline.

To activate it:

- [ ] Confirm the intended database environment.
- [ ] Map canonical transactions, fees, refunds, partners, and costs.
- [ ] Approve metric contracts and reporting periods.
- [ ] Implement a permission-safe aggregate adapter.
- [ ] Add lineage, watermark, completeness, and metric version.
- [ ] Connect the dashboard through its same-origin adapter.
- [ ] Validate filter consistency and refresh failure behavior.
- [ ] Deploy the alert worker with approved recipients and thresholds.
- [ ] Test deduplication, acknowledgment, recovery, and delivery failures.
- [ ] Backtest the forecast before publishing it as an operational forecast.

The financial model and code now support design review and implementation. Live data connection, deployment, and notification activation still require verified mappings and separately approved external changes.

⁂

1. The Main Semester design system (2) copy/handoff/specifications/role-system-execution-prompt.pdf
2. The Main Semester design system (2) copy/docs/system/ROLE_SYSTEM_RECONCILIATION.md